Year 12 Maths - Working with data

Similar questions - Interpreting the large data set

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The table gives information about the way employees travel to work in one Local Authority, taken from the large data set.

	2001	2011
All people in employment	40 000	48 000
Drive a car or van	24 000	26 400
Use public transport	3 200	5 280

(a) Calculate the percentage of employees who drove in 2001 and in 2011. **[2]**

(b) A councillor says: "Between 2001 and 2011 fewer people drove to work." Use the table to explain why this is misleading. **[2]**

(c) Find the percentage increase in the number of employees using public transport, and explain why this is not the same as the change in the percentage of employees using public transport. **[3]**

2 The table shows the age structure (percentage of residents in each age group) for four Local Authorities in 2011.

LA	0–17	18–24	25–64	65 and over
A	20	8	52	20
B	17	22	51	10
C	24	9	52	15
D	15	7	48	30

(a) Using only these columns, decide which LA is most likely to contain a large university, which is most likely to be attractive to retired people, and which is most likely to have many young families. Give a reason for each. **[3]**

(b) In a larger set of 18 LAs, the quartiles of the "65 and over" column are 14.6, 16.0 and 17.5 and the mean is 18.2. Explain whether the median or the mean is a better measure of a typical LA, and what this suggests about the distribution. **[3]**

(c) Explain why the table cannot be used to compare the *number* of retired people in LAs A and D. **[1]**

Question 1

(a) 2001: $\frac{24000}{40000} = 60\%$. 2011: $\frac{26400}{48000} = 55\%$

(b) The *percentage* driving fell, but the *number* driving increased by 2400 (from 24 000 to 26 400), because the total number of employees grew

(c) Percentage increase $= \frac{5280 - 3200}{3200} \times 100 = 65\%$. The percentage of employees using public transport rose from 8% to 11%, an increase of 3 percentage points. The two differ because the total number of employees also changed, and one compares numbers while the other compares proportions of different totals

Question 2

(a) University: B, largest percentage aged 18–24 (22%). Retired people: D, largest percentage aged 65 and over (30%). Young families: C, largest percentage aged 0–17 (24%)

(b) The mean (18.2) is greater than the median (16.0), which suggests a few LAs with unusually high percentages of older residents, so the distribution is positively skewed. The upper outlier limit is $17.5 + 1.5 \times 2.9 = 21.85$, so such values could be outliers. The median is the better measure of a typical LA

(c) Percentages are of different total populations; D could be much smaller than A, so the actual number of retired people could be fewer in D despite a higher percentage